

TerraBit

Low-Bit Quantization of Geospatial Embedding Lakes

Isaac Corley

Independent Researcher

ICML 2025 Workshop

Outline

Motivation

Method

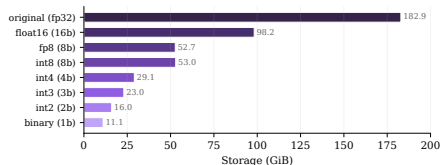
Results

Pipeline

Takeaways

The Embedding Lake Problem

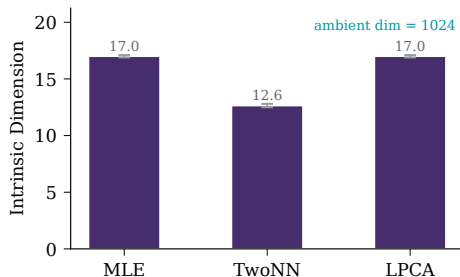
- Geospatial foundation models (Clay, Prithvi, GFM) produce planet-scale **embedding lakes**
- $50\text{M embeddings} \times 1024\text{D} \times \text{float32} = \mathbf{183\text{ GiB}}$
- At deployment: *storage and bandwidth* dominate cost, not model accuracy
- Current practice: `int8` (AlphaEarth, TESSERA)
- Open question: *how far below `int8` can we go?*



Intrinsic Dimension Diagnostics

Why aggressive quantization can work

- Before compressing: estimate the intrinsic dimension (ID) of the embedding manifold
- Three estimators: MLE, TwoNN, Local PCA
- Result: $\hat{d} \approx 13\text{--}17$ vs. ambient $d = 1024$
- Massive structural redundancy $\Rightarrow$ most per-dimension variance is irrelevant for local geometry
- Implication: aggressive quantization should not collapse neighborhood structure



Quantization Methods

Seven post-hoc methods from 16 bits to 1 bit

Standard Methods

- float16 — IEEE half-precision truncation
- fp8 E4M3 — 8-bit float, 4-bit exponent
- int8/4/3/2 — per-dim affine quantization with stored scale/zero-point, bit-packed for $b < 8$
- binary — sign-bit encoding, packed to $d/8$ bytes, Hamming-distance search

TurboQuant

- Random orthogonal pre-rotation R before quantization
- $\hat{x}_i = R^\top \mathcal{Q}_b(Rx_i)$
- Reduces inter-channel variance disparity
- Tested at $b \in \{2, 3, 4, 8\}$ bits
- Helps most at ultra-low bit-widths

Evaluation Approach

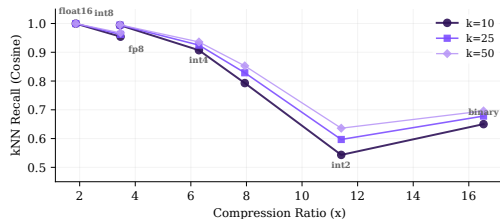
Label-free, retrieval-oriented metrics

- No labels needed — evaluation grounded in intrinsic-dimension analysis
- **kNN recall@ k** ($k \in \{10, 25, 50\}$) under cosine and Euclidean distance
 - Fraction of true k -nearest neighbors preserved post-quantization
- **Cosine rank correlation** — Spearman ρ of pairwise similarities (original vs. quantized)
- **Reconstruction cosine** and MSE (where dequantization available)
- Quality metrics on 20,000-row reservoir sample; reconstruction exact over all 49.8M rows

Quality–Compression Trade-off

Smooth degradation from float16 to binary

Method	Ratio	C@10	Cos Corr
float16	1.9×	1.00	1.00
fp8	3.5×	0.95	1.00
int8	3.5×	0.99	1.00
int4	6.3×	0.91	1.00
int3	7.9×	0.79	0.99
int2	11.4×	0.54	0.95
binary	16.5×	0.65	0.79

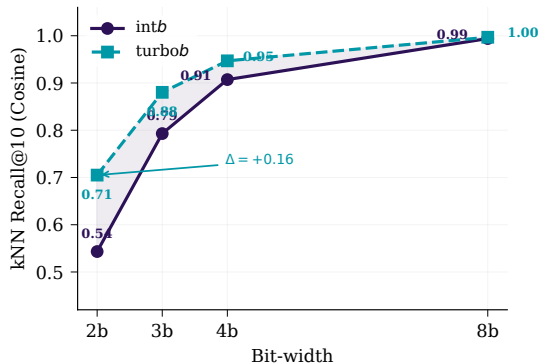


- int8: near-lossless ($R@10 = 0.99$)
- int4: sweet spot ($6.3\times$, 91%)
- binary: coarse retrieval ($16.5\times$, 65%)

TurboQuant vs. Standard Quantization

Rotation helps most at ultra-low bits

Method	C@10	Δ
int2	0.54	—
turbo2	0.71	+0.17
int3	0.79	—
turbo3	0.88	+0.09
int4	0.91	—
turbo4	0.95	+0.04
int8	0.99	—
turbo8	1.00	+0.01

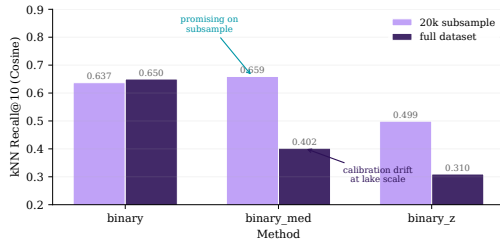


Rotation redistributes variance across channels $\Rightarrow$ better range utilization at coarse quantization. At 8 bits, affine quant is already near-lossless.

Binary Variants at Scale

Per-partition calibration fails at lake scale

- Tested three post-hoc binary improvements:
 - Per-dim median thresholding
 - Z-score thresholding
 - ITQ-style PCA rotation
- `binary_med` looks promising on subsample (recall 0.637 $\rightarrow$ 0.659)
- At full corpus: drops to **0.40** — worse than plain sign coding
- Root cause: per-file calibration drift across geohash/year partitions
- Takeaway: plain global sign coding is the strongest lake-scale binary baseline



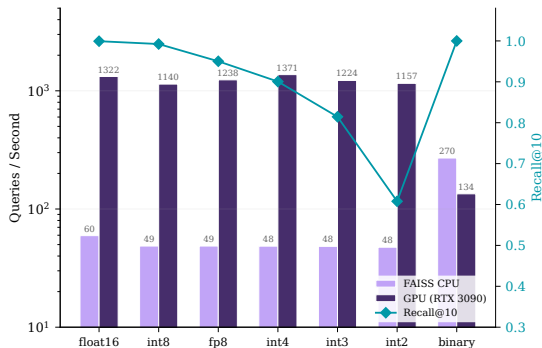
Search Throughput Benchmark

Brute-force kNN on 1M vectors (1K queries, $k=10$)

Method	CPU	GPU	R@10
float16	60	1322	1.00
int8	49	1140	0.99
int4	48	1371	0.90
int2	48	1157	0.61
binary	270	134	1.00*

* vs. binary GT (Hamming)

QPS = queries/sec. GPU = RTX 3090.
GPU: 22–28× speedup over FAISS CPU.



Transformation Pipeline

Schema-preserving parallel Parquet processing

- Hive-partitioned Parquet: 511 geohash cells $\times$ year/month
- Pipeline reads embedding column, applies quantizer, writes output preserving directory structure + metadata columns
- Quant params (scale, zero-point, method, bit-width) in Parquet footer metadata
- Full corpus (7 methods, 8 workers):
 - **45 min** wall-clock
 - 18,359 rows/s
 - 107 MiB/s output throughput

Design Principles

1. No retraining required
2. Preserves spatial/temporal metadata
3. Downstream-transparent: same partition layout
4. Self-describing: metadata in Parquet footer

Key Takeaways

1. **ID $\approx$ 13–17** vs. $d = 1024 \Rightarrow$ massive redundancy justifies aggressive quantization
2. **Smooth quality–cost frontier:** int4 at 91% recall / $6.3\times$ compression is the sweet spot
3. **Binary** ($16.5\times$) is viable for coarse-stage candidate generation in multi-stage retrieval
4. **TurboQuant** rotation helps most at 2–3 bits; negligible at ≥ 8 bits
5. **Per-partition calibration is a trap:** global transforms only for lake-scale binary
6. **GPU search:** > 1300 QPS on a single RTX 3090 with dequantized embeddings

Practical Guidance

Use int8 for near-lossless, int4 for balanced, binary for coarse retrieval. Add TurboQuant only at ≤ 3 bits.

Limitations & Future Work

Limitations

- Label-free proxies only — no task-specific evaluation (e.g. land-cover classification)
- Reservoir-sampled kNN metrics, not full-corpus exact
- Brute-force search only — no ANN index evaluation (IVF, HNSW)

Future Directions

- Labeled downstream evaluation
- ANN index $\times$ low-bit quantization interaction
- Stack post-hoc quantization on learned discrete codes (FSQ, VQ-VAE)
- Global calibration strategies for binary

Thank You

Questions?

Isaac Corley

isaac.corley@proton.me

<https://github.com/isaaccorley/terrabit>